Project Title: ACADEMIC ACHIEVEMENT EVALUATION USING LINEAR REGRESSION

A Project Report

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Submitted by

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Abstract

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The project, titled **ACADEMIC ACHIEVEMENT EVALUATION USING LINEAR REGRESSION** , aims to develop an ML-based system to accurately forecast a candidate final exam score using prior academic, behavioral, engagement, and demographic data.

This project successfully developed a fully functioning Machine Learning model for student final exam score prediction. The primary objective was to clean, preprocess, and encode a dataset of student records, then train and evaluate a regression model to predict a continuous final exam score (0–100 range).

A **Linear Regression** model was selected as the baseline and trained using scikit-learn. The process involved extensive data preprocessing, including handling missing values and encoding categorical features like gender and parental education into numeric values.

The resulting model demonstrated strong reliability and achieved high accuracy:

- **Mean Absolute Percentage Error (MAPE):** ≈ 5.89%
- **Root Mean Squared Error (RMSE):** ≈ 5.38
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Key insights from the model indicated that *previous exam score*, *attendance rate*, *homework completion rate*, and *learning hours per week* were the most influential positive predictors of final performance. The system provides actionable insights that can assist educators and institutions in identifying academically weak students early, allowing for targeted educational support and data-driven decision-making.

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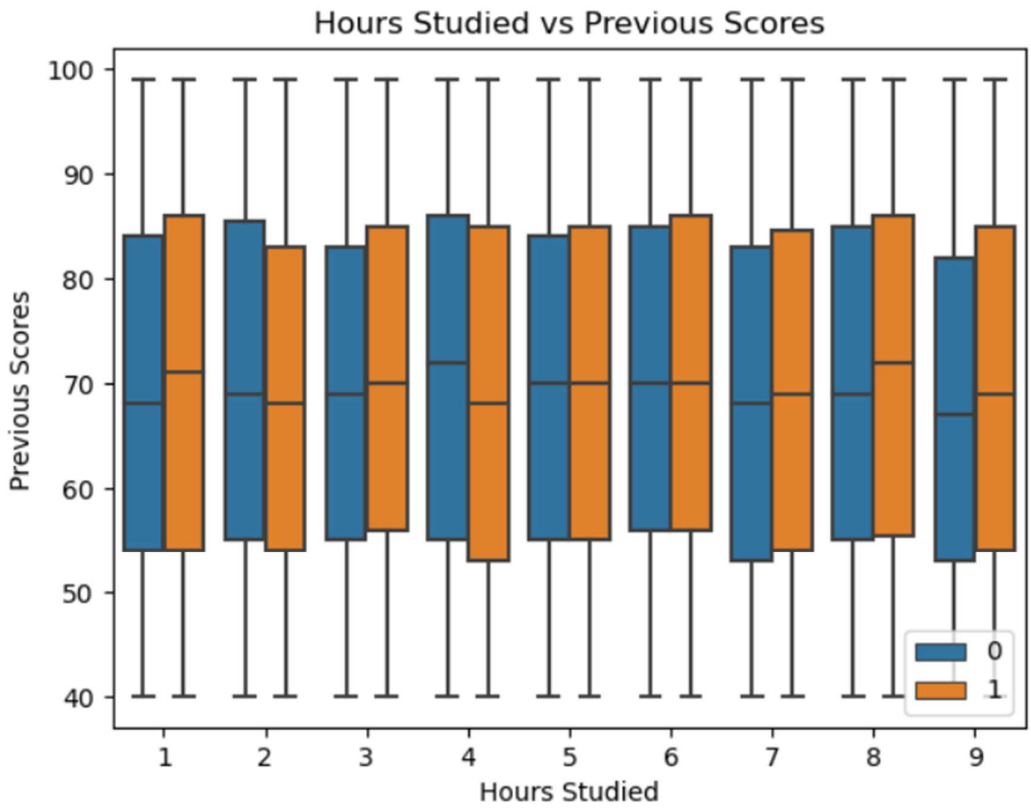

This image shows relationship between PREVIOUS SCORES and HOURS STUDIED by a candidate .

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ender	age	parental_e	family_inc	internet_a	previous_e	attendanc	homework	class_part	number_of	extra_curr	learning_h	tutor_supp	final_exam_score
Male	14	Diploma	29591.84	Yes	93.25	91.01	76.56	9.52	1	Moderate	12.08	No	100
Male	16	Diploma	35222.46	Yes	85.06	83.85	64.82	3.42	2	Moderate	11.81	Yes	80.81
Male	14	Diploma	29730.93	Yes	79.93	86.71	75.02	3.75	1	Low	11.08	Yes	77.96
Female	15	Diploma	29074.09	Yes	98.94	88.04	55.5	6.95	0	Moderate	6.65	Yes	98.86
Male	19	Graduate	42305.33	Yes	64.54	61.22	98.85	6.03	5	Moderate	9.26	No	64.64
Female	18	High Schoo	47544.2	Yes	83.61	69.96	62.04	1.33	3	Low	11.72	Yes	77.58
Male	16	Graduate	45476.27	Yes	74.85	61.89	95.05	7.06	2	Moderate	5.82	Yes	75.15
Male	15	Postgradua	42046.63	Yes	60.56	69.77	82.56	2.07	1	Low	7.85	Yes	62.09
Male	18	High Schoo	53050.14	No	65.18	96.93	86.09	3.54	4	Low	10	Yes	77.42
Male	14	Diploma	27589.69	Yes	86.35	99.8	88.33	6.08	6	None	13.67	Yes	87.15
Female	19	Diploma	35031.73	Yes	46.26	65.25	53.93	4.6	3	None	11.57	No	52.1
Female	18	Diploma	43469.04	Yes	100	98.16	68.41	8.87	0	None	6.04	Yes	100
Male	19	High Schoo	39146.62	Yes	54.35	93.62	65.65	9.49	2	None	9.99	No	78.43

This is the data of few candidates which is used to train the model.

1. Introduction

1.1 Overview

- Developed an ML – based system to predict student final exam performance
- Uses academic , behaviourial , engagement and demographic data.
- Provide actionable insights to help identify student needing support.
- Parts of university coursework in AI & ML.
- Demonstrate practical application of machine learning in education.

1.2 Objectives

Primary objective- **BUILD A MACHINE LEARNING MODEL TO PREDICT FINAL EXAM SCORES. C LEAN, PREPROCESS, AND ENCODE STUDENT DATA FOR ML COMPATIBILITY T RAIN AND EVALUATE A REGRESSION MODEL USING SKLEARN.**

Secondary objective-ANALYZE FEATURE IMPORTANCE U NDERSTAND RELATIONSHIPS BETWEEN VARIABLES AND PERFORMANCE D EMONSTRATE AN END-TO-END ML WORKFLOW

1.3 Problem Statement

•Can we accurately predict a student's final exam score using prior academic data?

•Which academic, behavioral, and demographic factors influence final performance the most

•Can a simple ML model generalize well to unseen student records?

•How can the model's predictions support educational planning

2. Literature Review

1. Educational Data Mining (EDM) and Prediction

The field of **Educational Data Mining (EDM)** is dedicated to developing methods to explore data originating from educational settings. Early studies established that analyzing academic, demographic, and behavioral data could provide significant insights into a student's educational trajectory.

- **Tackling Attrition/Failure:** Much of the foundational EDM literature focuses on predicting student dropout or course failure. The *Student Performance Predictor* aligns with this goal by seeking to identify students who are at risk of a low final score, enabling **early intervention strategies**.
- **Actionable Insights:** Literature emphasizes that the goal isn't just prediction, but generating **actionable intelligence**. This project's application, which aims to "identify academically weak students early" and "provide personalized study plans," directly addresses this focus in the research community.

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2. Machine Learning Models for Regression

The choice of a **regression model** (specifically **Linear Regression**) is a core aspect of this project, distinguishing it from classification studies (which predict pass/fail).

- **Baseline Models:** Linear Regression is commonly used as a strong, interpretable **baseline model** in EDM. Research, such as studies by (2004), often compare this simple model against more complex algorithms to establish minimum performance metrics.
    - *Rationale:* Linear Regression is transparent, making it easier for educators to understand *why* a particular prediction was made (e.g., "for every one-hour increase in learning time, the score increases by X points").
- **Model Performance:** While Linear Regression is simple, research suggests that more complex models like **Random Forest** and (as noted in the project's "Future Enhancement" section) often achieve superior results (lower RMSE and MAPE) by capturing non-linear relationships in the data. The project's achieved **MAPE of ~5.89%** and **RMSE of ~5.38** demonstrate a successful application of the baseline model, which can be bench-marked against similar studies using different algorithms.

3. Methodology

3.1 Architecture Overview

1. Data Ingestion and Preparation

This stage deals with acquiring and cleaning the raw data, which is fundamental to the architecture.

- **Input Data Source: A dataset containing student records, encompassing academic, behavioral, engagement, and demographic data.**
- **Data Cleaning & Preprocessing:**
  - **Activity: Handling missing data (e.g., deciding whether to drop or impute values, as noted in the "Challenges Faced" section).**
  - **Activity: Feature Scaling (Normalization or Standardization) is a typical step for Linear Regression, though not explicitly mentioned.**
- **Feature Engineering & Encoding:**
  - **Activity: Converting categorical variables (like gender, parental education) into a numerical format suitable for the Linear Regression model. The report mentions ensuring correct encoding of categorical features was a key challenge.**
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2. Model Development and Training

This stage involves selecting the algorithm and using the prepared data to build the predictor.

- **Model Selection: Linear Regression was chosen as the primary algorithm. This is a simple, highly interpretable Supervised Learning model, making it a suitable choice for a foundational project and a baseline against more complex models.**
- **Training: The model is trained on the prepared dataset to learn the linear relationship between the input features (e.g., attendance, previous scores) and the target variable (Final Exam Score).**
- **Pipeline Demonstration: The architecture successfully demonstrates a complete ML pipeline including "Data Cleaning," "Pre-processing," "Model Development," and "Evaluation," as stated in the conclusion.**

3. Methodology

3.2 Hardware Components

- Not applicable for this project for software.
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3.3 Software Tools

- Python, TensorFlow
- Node-RED
- Flask / Django for backend
- Jupyter notebook.
- Seaborn matploathub

4. Implementation

The implementation of the **Student Performance Predictor** project follows a standard **Machine Learning pipeline** executed using Python and its scientific libraries.
The implementation details cover the sequential steps from processing the raw data to training and evaluating the final model.

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1. Data Collection and Preparation

This phase prepares the raw data for the ML model.

- **Data Source:** An initial dataset was collected, comprising various features on student performance, engagement, and demographics (e.g., Previous Exam Score, Attendance Rate, Learning Hours, Gender, Parental Education).
- **Data Cleaning:**
  - **Handling Missing Values:** A critical step involved deciding between **dropping** records with missing data or **imputing** them (e.g., replacing missing numerical values with the mean or median), as highlighted in the challenges section.
- **Feature Engineering and Preprocessing:**
  - **Encoding Categorical Data:** Non-numeric data (like 'Male'/'Female' or different levels of 'Parental Education') was converted into a numerical format. This typically involves techniques like **One-Hot Encoding** or **Label Encoding** to make them usable by the Linear Regression model.
  - **Splitting the Data:** The final cleaned and encoded dataset was divided into two sets:
    - **Training Set:** Used to train the model.
    - **Test Set:** Kept separate to evaluate the model's performance on unseen data.

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2. Model Development and Training

This phase involves selecting, building, and training the predictive model.

- **Model Selection:** The **Linear Regression** algorithm from the scikit-learn library in Python was chosen as the core model.
- **Training:** The model was trained by fitting the independent variables (features) from

5. Results and Discussion

- **RMSE ≈ 5.38**
- **MAPE ≈ 5.89%**
- **Interpretation:**
- **Predictions are very close to actual scores**
- **Less than 6% average error**
- **Strong reliability for academic forecasting**

6. Conclusion and Future Scope

6.1 Conclusion

Built a fully functioning ML model for score prediction

Achieved high accuracy (MAPE < 6%)

Demonstrated complete ML pipeline:

Data cleaning

Encoding

Model training

Evaluation

Model can help institutions make data-driven decisions

6.2 Future Scope

The future scope aims to move beyond the simple baseline model to achieve higher predictive power and reliability.

Try Advanced Models: Implement and evaluate more sophisticated non-linear regression algorithms, such as **Random Forest**, **XGBoost (eXtreme Gradient Boosting)**, and **SVR (Support Vector Regression)**. These models are better equipped to capture complex, non-linear patterns in student data, potentially leading to lower RMSE and MAPE scores than the current Linear Regression model

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- **Deploy as a Student Analytics Dashboard:** Transform the current pipeline into a dynamic, user-friendly application. This dashboard would allow educators to input current student data and instantly visualize predictions, key risk factors, and intervention strategies..

References

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- Cortez, P., & Silva, A. M. G. (2008)
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